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Sequential Analysis of Variance: Increasing Efficiency of Hypothesis Testing

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Abstract

Researchers commonly use analysis of variance (ANOVA) to statistically test results of factorial designs. Performing an a-priori power analysis is crucial to ensure that the ANOVA is sufficiently powered, however, it often poses a challenge and can result in large sample sizes, especially if the expected effect size is small. Due to the high prevalence of small effect sizes in psychology, studies are frequently underpowered as it is often economically unfeasible to gather the necessary sample size for adequate Type II error control. Here, we present a more efficient alternative to the fixed ANOVA, the so-called sequential ANOVA that we implemented in the R package "sprtt". The sequential ANOVA is based on the sequential probability ratio test (SPRT) that uses a likelihood ratio as a test statistic and controls for long-term error rates. SPRTs gather evidence for both the null and the alternative hypothesis and conclude this process when a sufficient amount of evidence has been gathered to accept one of the two hypotheses. Through simulations, we show that the sequential ANOVA is more efficient than the fixed ANOVA and reliably controls long-term error rates. Additionally, robustness analyses revealed that the sequential and fixed ANOVA exhibit analogous properties when their underlying assumptions are violated. Taken together, our results demonstrate that the sequential ANOVA is an efficient alternative to fixed sample designs for hypothesis testing.

Keywords: sequential anova, sequential testing, sequential probability ratio tests

Word count: 6700

Sequential Analysis of Variance: Increasing Efficiency of Hypothesis Testing

When testing hypotheses, fixed-sample designs such as the Neyman-Pearson procedure (Neyman & Pearson, 1933) are the method of choice in psychological research. The Neyman-Pearson procedure allows researchers to decide between two competing statistical hypotheses with certain probabilities to make an erroneous decision. These probabilities are denoted by α, the probability to falsely reject a true null hypothesis H_0 (Type-I error), and β, the probability to falsely reject a true alternative hypothesis H_1 (Type-II error). While the former is controlled by choosing an appropriate critical value for the test statistic, the latter is controlled by collecting a sufficiently large sample size, which is determined by an *a priori* power analysis (Cohen, 1992; Erdfelder et al., 1996).

Even though this procedure is widely accepted as a nominal standard in psychology, *a priori* power analyses are not without challenges. First, researchers must specify an expected effect size, which may be difficult. If neither a precise prediction nor previously reported effect sizes are available to guide the specification, a reasonable approach is to use the smallest effect size of interest (Lakens, 2022). This strategy, however, is related to a second challenge: The required sample size for a statistical test is a function of the test procedure (e.g., parametric vs. non-parametric tests), the desired α (which is usually set to .05 or lower), the statistical power $1 - \beta$ (which is usually set to .80 or higher), and the hypotheses, that is, the expected effect size. When the effect size is small, the sample size required to ensure sufficiently low error probabilities is large. Unfortunately, psychological effects are typically small or medium sized (Funder & Ozer, 2019; Richard et al., 2003; Roberts et al., 2007). Thus, power analyses based on realistic effect-size expectations or smallest effect sizes of interest typically yield required sample sizes that exceed available resources. As a result, psychological studies are often based on too small sample sizes and thus underpowered (Button et al., 2013; Rossi, 1990; Szucs & Ioannidis, 2017).

A promising way to overcome the pervasive problem of underpowered studies due to

scarce resources is to resort to more efficient statistical testing procedures, namely, sequential hypothesis tests. Almost 80 years ago, Abraham Wald devised a sequential procedure to increase the efficiency of statistical decision-making in industrial production, the sequential probability ratio test (SPRT; Wald, 1945). Despite their favorable properties, the SPRT and other sequential designs have long been neglected in psychological research. Fostered by the increasing awareness of the need for efficient hypothesis-testing procedures and recent publications on the topic (e.g., Lakens, 2014; Pramanik et al., 2021; Schnuerch & Erdfelder, 2020; Schönbrodt et al., 2017; Stefan et al., 2022), however, there is a growing interest in sequential designs. With this article, we further support this development by implementing an SPRT for one of the most common test procedures, namely, the one-way analysis of variance (ANOVA). By thoroughly testing this sequential ANOVA in comparison to the conventional fixed-sample approach and by implementing it in a user-friendly R package, we hope to make the SPRT and sequential testing in general more accessible as a useful tool for psychological researchers in all subfields.

The Sequential Probality Ratio Test

Like the seminal theory of statistical decision making by Neyman and Pearson (Neyman & Pearson, 1933), the SPRT is a procedure for testing statistical hypotheses with controlled error probabilities (Wald, 1945; Wald, 1947). Unlike classical test procedures, however, the SPRT dispenses with the requirement of a fixed sample size (Wald, 1945). Instead, a decision is made after each observation whether the data collection can be terminated or needs to be continued. This flexibility to terminate early when there is compelling evidence in the data for one or the other hypothesis leads to an attractive feature: Compared to a fixed-sample design with the same error probabilities, the SPRT requires on average up to 50% smaller sample sizes (Schnuerch & Erdfelder, 2020; Wald, 1945). Thus, it is a highly efficient alternative to conventional testing procedures that control statistical decision errors.

The basic principle of the SPRT can be described as follows. The data are collected and monitored continuously. After each additional observation, a decision is made based on a predefined decision criterion whether one of the specified hypotheses (H_0 or H_1) can be accepted, in which case sampling is terminated. Otherwise, the data collection must be continued. Thus, the final sample size is a random variable that depends on the sequence of collected observations and the specified decision criterion.

Let us assume that we are interested in the random variable X, with $X \sim f(x;\ \theta)$, where θ denotes the true but unknown parameter vector of the underlying population and $f(.)$ may be either a probability mass function (if X is discrete) or a probability density function (if X is continuous). Further assume that we wish to test the two competing hypotheses H_0: $\theta = \theta_0$ versus H_1: $\theta = \theta_1$. The central component of the SPRT is the ratio of probabilities of the oberved data under the two hypotheses, that is, the likelihood ratio (LR). The LR is a measure of relative statistical evidence in the data for the specified hypotheses (Etz, 2018). Importantly, this interpretation is independent of how the data were sampled (Rouder, 2014). Starting at $n = 1$, the LR is calculated after any nth observation. Let $\mathbf{x}^{(n)} = (x_1, \ldots, x_n)$ be the vector of data at the nth stage. The LR at this stage is given by

$$LR_n = \frac{f(\mathbf{x}^{(n)};\ \theta_1)}{f(\mathbf{x}^{(n)};\ \theta_0)}. \tag{1}$$

Data collection continues by adding independent observations x_{n+1} as long as $B < LR_n < A$, where A and B denote predefined constants with $B < A$. As soon as $LR_n \geq A$, sampling is terminated and H_0 is rejected (H_1 is accepted); if, in contrast, $LR_n \leq B$, sampling is terminated and H_0 is accepted (H_1 is rejected). Even though the exact sample size is unknown beforehand, it was shown that this sequential procedure terminates with a probability of one after a finite number of observations (Wald, 1945).

To ensure that the test satisfies error probabilities α and β under H_0 and H_1,

respectively, appropriate values for A and B must be chosen. In practical applications, the choice is surprisingly simple: Wald (1945) showed that in order to achieve the desired error probabilities, upper and lower limits for A and B are given by $(1-\beta)/\alpha$ and $\beta/(1-\alpha)$, respectively. In practice precisely these limits may be used to define the threshold values, that is,

$$A = \frac{1-\beta}{\alpha} \tag{2}$$

and

$$B = \frac{\beta}{1-\alpha}, \tag{3}$$

and the resulting sequential procedure will have error probabilities smaller than or equal to α and β, respectively (Schnuerch et al., 2022; Schnuerch & Erdfelder, 2020; Wald, 1947).

The SPRT controls statistical decision errors in the same way as conventional test procedures with fixed sample sizes. In addition, it has the advantage of stopping early when there is compelling evidence in the data. In fact, Wald and Wolfowitz (1948) proved that the SPRT has an *optimum property*, that is, for given hypotheses and error rates, there is no (sequential or non-sequential) test procedure that requires a smaller sample size on average than the SPRT. This proof, however, only holds for *simple* hypotheses. A hypothesis is considered simple if the probability distribution is fully specified, that is, all parameters of the statistical model are either known or defined by the hypothesis. If, in contrast, there is at least one free parameter, the hypothesis is *composite* (Neyman & Pearson, 1933). For example, when testing hypotheses about the mean of a normally distributed variable with unknown population variance σ^2, the hypotheses are composite as

long as they do not specify the value of σ^2. In psychological research, population variances are typically unknown, which limits the applicability of the original SPRT as outlined above. However, extensions of the SPRT have solved this issue for a range of testing scenarios, for example, sequential t-tests (Hajnal, 1961; Rushton, 1950). Even though the proof of the SPRT's optimum property does not necessarily hold for these scenarios, the sequential tests have been shown to still be much more efficient than corresponding fixed-sample tests (e.g., Schnuerch & Erdfelder, 2020).

Despite their efficiency and flexibility, SPRTs are not very popular in psychology (Ulrich & Miller, 2021). One of the reasons for the widespread neglect of sequential procedures may be that, first, most of the current literature on the SPRT focuses on t-tests and not on more complex applications (e.g., Schnuerch & Erdfelder, 2020; Stefan et al., 2022; but see Schnuerch et al., 2020, 2022). Second, SPRTs exist mainly in the literature but not in easily applicable, well-documented, and flexible statistics software (Albers, 2019).

In order to address the first problem and improve the applicability of SPRTs, we implemented a sequential one-way fixed-effects ANOVA that allows researchers to perform efficient hypothesis tests on multiple means instead of only two. In the present study, we investigate through simulations its properties in direct comparison to a fixed-sample ANOVA and demonstrate that the sequential ANOVA controls α and β notably more efficiently. We also report the procedures' performances under imperfect conditions such as unequal sample sizes or violations of homoscedasticity. Finally, we illustrate and discuss a limitation of sequential ANOVAs, namely, biased effect-size estimation following a sequential test.

In order to address the second problem, we developed the *sprtt* (Sequential Probability Ratio Test Toolbox) package in R, which contains implementations of the sequential ANOVA and sequential t-tests (Steinhilber et al., 2023). The package was

published on CRAN and the latest development version can be found on GitHub: https://github.com/MeikeSteinhilber/sprtt. The implementation was done carefully in consideration of future extendability and maintenance requirements. This user-friendly package makes applications of the SPRT straightforward and convenient for applied researchers. We hope that addressing these two issues – implementing a sequential one-way fixed-effects ANOVA and making SPRTs easily available in an R package – will contribute to the growing popularity of sequential analysis in psychological research.

Alternative Sequential Designs

A sequential procedure that is closely related to the SPRT is the so-called Sequential Bayes Factor Test (SBFT; Schönbrodt et al., 2017). The SBFT uses Bayes factors instead of likelihood ratios as a test statistic. To calculate Bayes factors, prior distributions are defined for unknown parameters and the likelihood is then integrated (*marginalized*) across these priors (Morey & Rouder, 2011; Wagenmakers et al., 2010). Thus, the Bayes factor can be understood as a weighted average of likelihoods with priors as weights (Rouder et al., 2009). Using a prior distribution instead of a point hypothesis has the advantage that researchers do not have to commit to a single effect size but that they can specify a range of plausible values. As has been shown in the context of sequential t-tests, this can decrease the efficiency especially for small to medium effect sizes, but it increases the robustness against model misspecifications (i.e., when the true effect size differs from the expected effect size; see Schnuerch & Erdfelder, 2020; Stefan et al., 2022). In contrast, SPRTs are on average more efficient and allow for the explicit control of error rates, but they are more sensitive to model misspecifications (Schnuerch & Erdfelder, 2020; Stefan et al., 2022). In the end, both methods are very similar and the decision to use either an SPRT with point hypotheses or an SBFT with prior distributions likely reflects a general preference for one statistical theory (i.e., frequentism vs. Bayesianism) over the other. As has recently been shown, the two procedures can even be integrated to combine

the benefits of both into a single testing framework (Schnuerch et al., 2022).

Other sequential procedures such as the Group Sequential Design (GSD; Lang, 2017; Proschan et al., 2006) or the Independent Segments Procedure (ISP; Miller & Ulrich, 2021) differ conceptually more from SPRTs than the SBFT. SPRTs allow for termination after any single observation while the final sample size is unknown beforehand (i.e., it is a *non-truncated* procedure). The ISP, in contrast, uses *a priori* planned interim analyses of batches of observations up to a precisely defined maximum (i.e., a *truncated* procedure). This sampling plan allows the analyst to stop data collection only at these planned stops. Although the ISP is more efficient than corresponding fixed-sample designs, due to this reduced flexibility in the sampling plan, it is on average less efficient compared to SPRTs (Erdfelder & Schnuerch, 2021).

GSDs are a family of test designs that allow for interim analyses (*looks*) and early termination based on different approaches to control error probabilities that vary in their complexity and flexibility (Lakens et al., 2021). Simpler but also less flexible versions use procedures such as the Pocock or O'Brien and Fleming procedure to correct the critical values for early stopping at each look (Lakens et al., 2021; O'Brien & Fleming, 1979; Pocock, 1977). However, these corrections can only be used if the number of looks as well as the required sample size for each look is known beforehand. More flexible alternatives use so-called *spending functions* that allow the analyst to control how the α and β rates are spent across looks while the exact number of looks and sample sizes are not required to be specified in advance (Lakens et al., 2021; Lan & DeMets, 1983).

One of the main advantages of GSDs and the ISP over non-truncated designs such as the SPRT is that $N_{\max}$ is known beforehand, which may be important for researchers when planning a study (Miller & Ulrich, 2021; Ulrich & Miller, 2021). In comparison to these truncated designs, however, SPRTs are more efficient on average. Moreover, as SPRTs are based on likelihood ratios, sampling can be stopped and the test statistic

(LR_N) can be interpreted as a direct measure of statistical evidence at any point – unlike a p value (Schnuerch & Erdfelder, 2020). Therefore, as long as $N_{\max}$ does not have to be known precisely in advance, SPRTs are a maximally efficient alternative to conventional fixed-sample procedures.

Sequential ANOVA

We implemented the SPRT for a one-way fixed-effects ANOVA using the same strategy as for the sequential probability ratio t-test (Schnuerch & Erdfelder, 2020): Let $\mathbf{X}^{(n)}$ be the observed data at the nth step of the data collection for k groups,

$$\mathbf{X}^{(n)} = \{\mathbf{x}_1 = (x_{1,1}, ..., x_{1,n}), \mathbf{x}_2 = (x_{2,1}, ..., x_{2,n}), ..., \mathbf{x}_k = (x_{k,1}, ..., x_{k,n})\}, \tag{4}$$

with the total sample size of $N = k \cdot n$, and $n, k \geq 2$. The likelihood ratio LR_n is defined as the ratio of the probability (density) of $\mathbf{X}^{(n)}$ under each hypothesis (Equation 5a). Without explicitly specifying the population means $\mu_1, \mu_2, \dots, \mu_k$ and the common population variance σ^2, however, this probability is undefined. As a remedy, the sequence of observed data may be transformed to the sequence of corresponding F-values whose distribution no longer depends on the unknown parameters, see Equation 5b (Barnard, 1949; Wetherill & Glazebrook, 1986). Following Cox's theorem (Cox, 1952), LR_n can then be expressed as the ratio of the probability densities of the most current F-value at each step n (F_n) under a non-central versus a central F-distribution, representing H_1 and H_0, respectively (Equation 5c):

$$LR_n = \frac{f(\mathbf{X}^{(n)};\ H_1)}{f(\mathbf{X}^{(n)};\ H_0)} \tag{5a}$$

$$= \frac{f(F_2, F_3, ..., F_n;\ H_1)}{f(F_2, F_3, ..., F_n;\ H_0)} \tag{5b}$$

$$= \frac{f(F_n;\ df_1, df_{2n}, \Delta_n)}{f(F_n;\ df_1, df_{2n})}. \tag{5c}$$

The F-distribution is defined by three parameters: Two degrees-of-freedom parameters (df_1 and df_2) and a non-centrality parameter (Δ). At each step n, the degrees of freedom are calculated with the well-known formulas using the sample size and the number of groups,

$$\begin{aligned} df_1 &= k - 1, \\ df_{2,n} &= N - k. \end{aligned} \tag{6}$$

The non-centrality parameter Δ is a function of the expected effect size and the sample size:

$$\Delta_n = f_{exp}^2 \cdot N, \tag{7}$$

where f_{exp} denotes the expected effect size in terms of Cohen's f and N is the total sample size at the nth step. Cohen's f is defined as: $f = \frac{\sigma_m}{\sigma}$ with $\hat{\sigma}_m = \sqrt{\frac{\sum_{i=1}^{k}(m_i - \bar{m})^2}{k}}$, where $\bar{m}_i$ is the mean of the group i and $\bar{m}$ is the overall mean of all k groups. Cohen's f measures the spread of the means (using the standard deviation of the means) divided by the common population standard deviation. Put differently, f is the standard deviation of standardized means if the data are standardized using the within-population standard deviation (Cohen, 2009, p. 275).

Expressed in terms of Cohen's f, the statistical hypotheses underlying the one-way ANOVA are

$$\begin{aligned} H_0 &: f = 0, \\ H_1 &: f = f_{exp} \ (f_{exp} > 0). \end{aligned} \tag{8}$$

Consequently, the F-distribution under the null hypothesis reduces to a central F-distribution with $\Delta = 0$, and the LR_n may be expressed as the ratio of the probability density of F_n under a non-central to a central distribution. Starting with $n \geq 2$[1], F_n is calculated as follows (Wetherill & Glazebrook, 1986, p. 81):

$$\begin{aligned} F_n &= \frac{SS_{\text{effect},n}/df_1}{SS_{\text{residual},n}/df_{2,n}}, \\ SS_{\text{effect},n} &= \sum_{i=1}^{k} n(\bar{x}_i - \bar{x})^2, \\ SS_{\text{residual},n} &= \sum_{i=1}^{k}\sum_{j=1}^{n}(x_{i,j} - \bar{x}_i)^2, \end{aligned} \tag{9}$$

where $\bar{x}$ is the overall mean, x_i is the mean of the group i, $x_{i,j}$ is the data point of the i-th group and the j-th position of n sequential steps, SS_{effect} denotes the effect sum of squares and $SS_{\text{residuals}}$ denotes the residual sum of squares.

The SPRT is then carried out by increasing the sample size as long as

$$\frac{\beta}{1-\alpha} < \frac{f(F_n;\ df_1, df_{2n}, \Delta_n)}{f(F_n;\ df_1, df_{2n})} < \frac{1-\beta}{\alpha}, \tag{10}$$

and terminates to accept H_0 (and reject H_1) as soon as the left inequality is violated, or terminates to accept H_1 (and reject H_0) as soon as the right inequality is violated.

For the case of simple hypotheses, the SPRT has well-defined properties that may be described analytically (Wald, 1947). When extended to account for composite hypotheses, however, the exact properties cannot be determined analytically. While there has been a number of simulation studies recently investigating the properties of sequential

[1] We assume in the formulas that in every sequential step one data point is added to each group. Consequently, the sample size of every group is equal to n.

t-tests, little is known about the sequential ANOVA's efficiency compared to the fixed-sample case, or about the test's sensitivity to violations of its underlying assumptions (e.g., Schnuerch & Erdfelder, 2020; Stefan et al., 2022). In this simulation study, we seek to fill this gap. In a first set of simulations, we investigate error rates and efficiency of the sequential ANOVA compared to the corresponding fixed ANOVA under ideal conditions, that is, when all assumptions underlying both tests are met. Subsequently, we look at how these properties are affected by imperfect conditions, namely, unequal group sizes and/or population variances, as well as violations of normality assumptions. Lastly, we also assess potential biases of effect size estimation following a sequential ANOVA. All simulation scripts, the simulation results and 1% of the simulated raw data can be found at the Open Science Framework (https://osf.io/6yq5f/).

Simulation Under Perfect Conditions

Settings of the Simulation

In every iteration of the simulation, we randomly drew 4 population data sets ($N_1 = N_2 = N_3 = N_4 = 20{,}000$) with a certain effect structure from normal distributions to simulate a study with 4 independent groups. The population variances (σ_i^2) were specified as 1. To vary the population means (μ_i), we drew 4 raw means (m_i) from a standard normal distribution. These raw means were then transformed to match a specified effect size of Cohen's f (f_{true}) for $k = 4$ groups. In the transformation, the raw means were z-standardized, scaled by f_{true}, and corrected by $\sqrt{\frac{k}{k-1}}$:

$$\mu_i = \frac{m_i - \bar{m}}{s_m} \cdot f_{true} \cdot \sqrt{\frac{k}{k-1}}, \; i \; = \; 1, 2, ..., k. \tag{11}$$

Then, we calculated a sequential and a fixed one-way ANOVA on samples from these simulated populations. The sequential ANOVA started with 2 data points in every group (i.e., $k \times 2 = 8$ data points in total), and one data point per group was added in every sequential step. The sequential data collection process was terminated as soon as the

upper or lower threshold value for LR_n was reached and a decision for one of the two hypotheses (H_0 or H_1) was made. In contrast, the fixed ANOVA was performed on a sample with a fixed size that was determined by an *a priori* power analysis. We used an α and β level of .05 for the power analysis and the specification of the sequential ANOVA. Altogether, 12 combinations of true and expected effect sizes were simulated ($f_{true} = \{0, 0.1, 0.25, 0.4\}$, $f_{exp} = \{0.1, 0.25, 0.4\}$). Each combination was repeated 10,000 times (N_r). Thus, our results are based on 120,000 simulated cases. These are the simulation settings for our *main simulation*, which we will reference later.

We used seeds for all simulations and stored 1% of the raw data. The seed for an iteration is the sum of a hyper seed (we used 10,000) and the number of the iteration. Therefore it was possible to get reproducible results while using parallelization to speed up the run time. All results were conducted multiple times with different seeds and the choice of seeds did not affect the pattern of results in a meaningful way.

Error Rates

Table 1 shows the simulated error rates of the sequential and the fixed ANOVA. To account for sampling error in the estimated rates, we calculated a 90% Clopper-Pearson interval as a criterion for inflated error rates. If the lower bound of the interval was equal to or larger than .05, the error rate was considered inflated. As expected, the sequential ANOVA controls the α and β error rates as long as the expected effect size is smaller than or equal to the true effect size in the population. Only if the true effect in the data is smaller than expected, the β error rates exceed the nominal value. To understand the increasing β error rates better, we investigated the error rates of a small effect ($f_{true} = 0.10$) while we slightly increased (+ 0.01) the expected effect size, starting from a perfect specification to a medium effect size ($f_{exp} = \{0.10, 0.11, 0.12..., 0.25\}$). Unsurprisingly, the results highlight that even a small overestimation of the true effect size negatively impacts Type-II error rates, for sequential and fixed ANOVAs alike (see Figure

1).

Table 1

Error Rates of the Sequential and Fixed ANOVA (4 groups, $\alpha = \beta = .05$, N_r = 10,000)

Error Type	f_{true}	f_{exp}	Error Rate Seq (%)[a]	Error Rate Fix (%)[b]
Type I	**0.00**	0.10	4.6	5.1
		0.25	3.8	5.1
		0.40	3.6	5
Type II	**0.10**	0.10	4.8	5
		0.25	*77.5*	*74.7*
		0.40	*91*	*88.1*
	0.25	0.10	0	0
		0.25	4	4.9
		0.40	*39.8*	*41.8*
	0.40	0.10	0	0
		0.25	0	0
		0.40	3.3	4.5

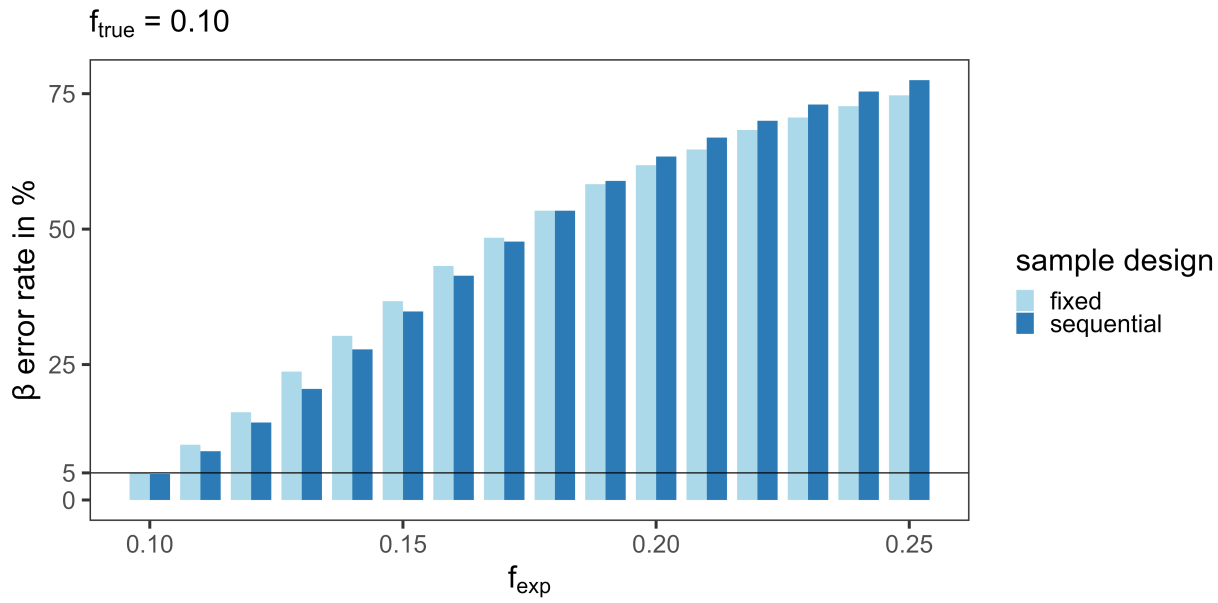
Note: The red color (and the italic font) indicates that the error rate is significantly higher than the alpha level of 5% using the lower boundary of the 90% Clopper-Pearson-Interval as criterion; f_{true} = true effect size in the population; f_{exp} = expected effect size; [a] error rate of the sequential ANOVA; [b] error rate of the fixed ANOVA; N_r = iterations per condition in the simualtion

Sample Size and Efficiency

To compare the efficiency of the sequential with the fixed ANOVA, the analyzed samples were always selected from the beginning of the simulated data vector; thereby, both procedures technically have the chance to use exactly the same data. Across

Figure 1

Error Rates of the Sequential and Fixed ANOVA (4 groups, $\alpha = \beta = .05$, N_r = 10,000)



Note: N_r = iterations per condition in the simualtion; f_{true} = true effect size in the population; f_{exp} = expected effect size

simulation conditions, the sequential procedure terminated with a sample size smaller than the fixed-sample procedure in 87% of the cases. On average, the sequential ANOVA required 58% smaller samples than the conventional ANOVA. In the following, we will refer to this reduction in the required sample size as *sample size efficiency*. To summarize, the sequential ANOVA is overall clearly more efficient than the fixed ANOVA. The exact sample size efficiency depends on the effect size, however (see Table 2): The smaller the effect size, the higher the sample size efficiency of a sequential compared to a fixed-sample ANOVA.

Table 2

Sample Size Efficiency of a Sequential ANOVA Compared to a Fixed ANOVA (4 groups, $\alpha = \beta = .05$, N_r = 10,000)

f_{exp}	Sample Size Efficiency (%)
0.10	64
0.25	40
0.40	23

Note: f_{exp} = expected effect size

To better understand this pattern, it is helpful to look at the average sequential sample sizes necessary to reach a decision compared to the fixed sample sizes. This comparison can be seen in Table 3. Especially the combination of $f_{exp} = 0.10$ and $f_{true} = 0.40$, where the true effect in the data was larger than expected, shows a striking difference in sample sizes. While the sample size of a sequential procedure is influenced by both, the *expected* effect size and the *true* effect size underlying the observed data, the sample size of a fixed design is solely based on the expected effect size. Even when there is overwhelming evidence in the data, the procedure does not allow for early termination. The sequential design, in contrast, automatically adapts to the stronger effect and, thus, requires notably smaller sample sizes.

Table 3

Required Sample Sizes (4 groups, $\alpha = \beta = 0.05, N_r$ = 10,000)

f_{exp}	f_{true}	Fixed Sample[a]	Mean Sequential Sample[b]
0.10	0.00	1724	1103
	0.10	1724	963
	0.25	1724	274
	0.40	1724	166
0.25	0.00	280	185
	0.10	280	235
	0.25	280	165
	0.40	280	86
0.40	0.00	112	77
	0.10	112	87
	0.25	112	108
	0.40	112	71

Note:

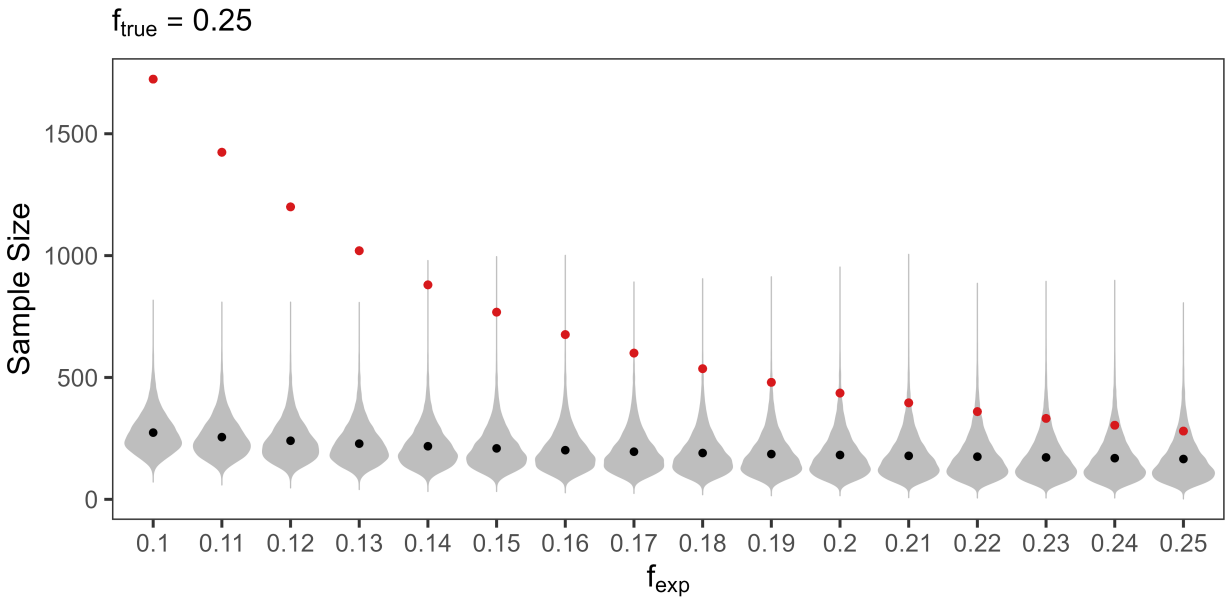
f_{exp} = expected effect size; f_{true} = true effect size in the population; [a] result of an *a priori* power analysis using the *pwr* package in R; [b] average sample size of sequential ANOVAs on simulated data; N_r = iterations per condition in the simualtion

Subsequently, we investigated the (average) sample sizes for a medium effect ($f_{true} = 0.25$) while we slightly increased (+ 0.01) the expected effect size, starting from a small effect up to a medium effect ($f_{exp} = \{0.10, 0.11, 0.12..., 0.25\}$). Figure 2 shows that underestimation of the true effect size ($f_{true} > f_{exp}$) results in higher average sample sizes for both test procedures, but the fixed procedure in particular shows a substantial increase

in the required sample size. In addition, Figure 2 highlights the variability of the sequential sample sizes and illustrates that, in some cases, the sequential sample size may even be larger than the fixed sample. This happens only rarely, however, such that in most cases, the sequential sample was noticeably smaller than the fixed sample.

Figure 2

Sample Sizes of Sequential and Fixed ANOVAs (4 groups, $\alpha = \beta = .05$, N_r = 10,000)



Note: N_r = iterations per condition in the simualtion; The red dots represent the results of the *a priori* power analysis of the fixed ANOVAs. The black dots represent the mean samle sizes that were necessary to reach a decision (H_1 or H_0) for the sequential ANOVAs.

Simulations of Imperfect Conditions

Examining statistical procedures under ideal conditions is a necessary step to understand the test properties. However, since empirical research is rarely conducted under ideal circumstances, it is essential to study the properties of sequential ANOVAs in scenarios where certain assumptions are violated. We examined 4 scenarios in more detail: (1) unequal group sample sizes, (2) unequal variances across groups, (3) unequal group sample sizes and variances, (4) and non-normally distributed data. This selection of

scenarios is based on common scenarios of robustness analysis of sequential t-tests and fixed ANOVAs (Blanca et al., 2018; Glass et al., 1972; Lee & Fung, 1980; Lix et al., 1996; Schnuerch & Erdfelder, 2020).

Unequal Group Sizes and Variances

Settings of the Simulation

In every iteration of the simulation, we randomly drew 4 population data sets with equal ($N_1 = N_2 = N_3 = N_4 = 20{,}000$) or unequal ($N_1 = N_2 \neq N_3 = N_4$) group sample sizes, with a total sample size of $N = 80,000$ and with a certain effect structure from normal distributions to simulate a study with 4 independent groups. The ratios between the samples were chosen as (1,1,1,1), (1,1,2,2), and (2,2,1,1), meaning that in the unbalanced cases, the two largest groups had twice the amount of data at each sequential step than the two smallest groups.

The population standard deviations were either held constant across groups (i.e., all $\sigma_i = 1$, equal condition) or simulated as $\sigma_1 = \sigma_2 = 1 \neq \sigma_3 = \sigma_4 = 2$ (unequal condition). To vary the population means (μ_i) in the groups, we drew 4 raw means (m_i) from a standard normal distribution. These raw means were then transformed to match a specified effect size of Cohen's f (f_{true}) for k groups:

$$\mu_i = \frac{m_i - \bar{m}}{s_m} \cdot f_{true.adj} \cdot \sqrt{\frac{k}{k-1}}, \; i \; = \; 1, 2, ..., k, \tag{12}$$

with $f_{true.adj} = f_{true} \cdot \sqrt{\frac{\sum_{i=1}^{k} \sigma_i^2 (n_i - 1)}{N-k}}$ in order to adjust for unequal variances. Then, we calculated a sequential and a fixed ANOVA on samples from these simulated populations. The sequential ANOVA started with at least two data points in every group, and one or two data points per group were added in every sequential step depending on the sample ratio. The sequential data collection process was terminated as soon as a decision for one of the two hypotheses (H_0 or H_1) was made. In contrast, the fixed ANOVA used a

sample with a fixed size, which was determined by an *a priori* power analysis assuming equal sample sizes.

We set $\alpha = \beta = .05$ for the power analysis and the specification of the sequential ANOVA. Altogether, 12 combinations of simulated and expected effect sizes were simulated ($f_{true} = \{0, 0.1, 0.25, 0.4\},\ f_{exp} = \{0.1, 0.25, 0.4\}$). Each combination was repeated 10,000 times.

Results

Table 4

Error Rates of the Sequential and Fixed ANOVA (4 groups, $\alpha = \beta = .05$, N_r = 10,000)

sd	Sample Ratio	Error Type	f_{true}	f_{exp}	Error Rate Seq (%)[a]	Error Rate Fix (%)[b]	Mean Sequential Sample[c]	Fixed Sample[d]
1,1,1,1	**1,1,2,2**	**Type I**	**0.00**	0.10	4.5	4.9	1108	1724
				0.25	3.8	5.4	190	280
				0.40	3.7	4.9	79	112
		Type II	**0.10**	0.10	*6.7*	*7.3*	1003	1724
			0.25	0.10	0	0	287	1724
				0.25	*6*	*7.2*	174	280
			0.40	0.10	0	0	173	1724
				0.25	0	0	92	280
				0.40	5.2	*7.2*	76	112
1,1,2,2	**1,1,1,1**	**Type I**	**0.00**	0.10	*7.5*	*6.1*	1035	1724
				0.25	*6.4*	*6.2*	175	280
				0.40	*6.2*	*6.1*	73	112
		Type II	**0.10**	0.10	4.1	4.6	958	1724
			0.25	0.10	0	0	274	1724
				0.25	3.9	4.8	165	280
			0.40	0.10	0	0	166	1724
				0.25	0	0	85	280
				0.40	3.1	4.3	70	112

Note: The red color (and the italic font) indicates that the error rate is significantly higher than the alpha level of 5% using the lower boundary of the 90% Clopper-Pearson-Interval as criterion; f_{true} = true effect size in the population; f_{exp} = expected effect size; [a] error rate of the sequential ANOVA; [b] error rate of the fixed ANOVA; [c] average sample size of sequential ANOVAs on simulated data; [d] result of an *a priori* power analysis using the *pwr* package in R; N_r = iterations per condition in the simualtion; The table includes only the cases where we have previously shown that the error rates were successfully controlled under ideal conditions ($f_{exp} \leq f_{true}$).

Table 5

Error Rates of the Sequential and Fixed ANOVA (4 groups, $\alpha = \beta = .05$, N_r = 10,000)

sd	Sample Ratio	Error Type	f_{true}	f_{exp}	Error Rate Seq (%)[a]	Error Rate Fix (%)[b]	Mean Sequential Sample[c]	Fixed Sample[d]
1,1,2,2	**1,1,2,2**	**Type I**	**0.00**	0.10	2.9	2.8	989	1724
				0.25	2.5	3.1	168	280
				0.40	2.3	2.9	70	112
		Type II	**0.10**	0.10	3.2	5	1121	1724
			0.25	0.10	0	0	291	1724
				0.25	3	5.2	194	280
			0.40	0.10	0	0	174	1724
				0.25	0	0	93	280
				0.40	2.7	5.2	82	112
	2,2,1,1	**Type I**	**0.00**	0.10	*18*	*12.4*	1040	1724
				0.25	*17*	*13.1*	182	280
				0.40	*15.3*	*13.7*	78	112
		Type II	**0.10**	0.10	*7.9*	*8.4*	880	1724
			0.25	0.10	0	0	280	1724
				0.25	*7*	*8.2*	157	280
			0.40	0.10	0	0	171	1724
				0.25	0.3	0.1	88	280
				0.40	*6.3*	*8*	68	112

Note: The red color (and the italic font) indicates that the error rate is significantly higher than the alpha level of 5% using the lower boundary of the 90% Clopper-Pearson-Interval as criterion; f_{true} = true effect size in the population; f_{exp} = expected effect size; [a] error rate of the sequential ANOVA; [b] error rate of the fixed ANOVA; [c] average sample size of sequential ANOVAs on simulated data; [d] result of an *a priori* power analysis using the *pwr* package in R; N_r = iterations per condition in the simualtion; The table includes only the cases where we have previously shown that the error rates were successfully controlled under ideal conditions ($f_{exp} \leq f_{true}$).

The error rates and required sample sizes of the sequential and the fixed ANOVA can be seen in Tables 4 and 5. Table 4 displays the cases where either the variance or the sample sizes in the groups vary, while Table 5 shows the cases where both variance and sample size differ between groups. If only the sample size ratio was unbalanced, the empirical β was increased by a maximum of 2.3 percentage points (see Table 4). If only the variance was unbalanced between groups, the empirical α was increased by a maximum of 2.5 percentage points.

When both the sample ratio and population variances were unbalanced, two scenarios could be distinguished: The population variance could be larger in groups with larger sample sizes, or it could be larger in groups with smaller sample sizes. As the first case may lead to an overall overestimation of the within-groups variance and, thus, an underestimation of the effect size, we did not observe an increase in error rates in this case. In the second case, however, the effect size is overestimated due to an underestimated within-groups variance. Consequently, error rates were inflated in this case. Specifically, we observed an increase in α by a maximum of 13 percentage points and an increase in β by a maximum of 3.4 percentage points, see Table 5.

Overall, the sequential and the fixed ANOVA showed similar error rates when assumptions about group sizes and homoscedasticity were violated. However, the sequential ANOVA was still more efficient than the fixed-sample procedure. The mean sequential sample sizes required by the sequential procedure were consistently lower than the corresponding fixed sample sizes.

Deviation from Normal Distribution

Mixture distributions are commonly used to simulate non-normal data for robustness analysis (Lee & Fung, 1980; Lix et al., 1996; Schnuerch & Erdfelder, 2020). Thus, we simulated data drawn from Gaussian mixture distributions to investigate the impact of normality-assumption violations on error rates and sample size efficiency of

sequential ANOVAs in comparison to fixed ANOVAs.

Settings of the Simulation

In every iteration of the simulation, we randomly drew 4 population data sets ($N_1 = N_2 = N_3 = N_4 = 20{,}000$) with a certain effect structure from Gaussian mixture distributions to simulate a study with 4 independent groups. The population standard deviations (σ_i) were specified as 1. To vary the population means (μ_i) in the groups, we drew 4 raw means (m_i) from a standard normal distribution. These raw means were then transformed to match a specified effect size of Cohen's f (f_{true}) for k groups based on the same formula as used in the main simulation (Equation 11).

We used a Gaussian mixture [$f(x) = p \cdot g_1(x) + (1 - p) \cdot g_2(x)$] with two normal distributions [$g_1 : Z_1 \sim N(\mu_{z.1}, \sigma_{z.1})$ and $g_2 : Z_2 \sim N(\mu_{z.2}, \sigma_{z.2})$] to simulate non-normal data (X) for every group with $X = I \cdot Z_1 + (1 - I) \cdot Z_2$, where Z_1 and Z_2 represent the data of the sub-populations and where

$$I = \begin{cases} 1 & \text{with probability } p, \\ 0 & \text{with } 1 - p \end{cases} . \tag{13}$$

We let $p = .50$, such that

$$E(Z) = \frac{(\mu_{z.1} + \mu_{z.2})}{2}, \tag{14}$$

and

$$Var(Z) = \frac{1}{2}(\sigma_{z.1}^2 + \sigma_{z.2}^2) + \frac{1}{4}(\mu_{z.1} - \mu_{z.2}). \tag{15}$$

We randomly generated values for the sub-population parameters using two constraints: (1) The resulting mean and variance of the mixture distribution must be equal

to certain predefined values, and (2) the mixture distribution should differ as much as possible from a normal distribution.

To generate $\mu_{z.1}$ and $\mu_{z.2}$, we used the information that the sum of these means equals $2E(Z)$. Thus, we drew the sub-population means as follows:

$$\begin{aligned} &\mu_{z.1} = 2E(Z) \cdot h_1, \\ &\mu_{z.2} = 2E(Z) \cdot (1 - h_1), \\ &\text{with } h_1 \sim U[0.8, 1]. \end{aligned} \tag{16}$$

To generate $\sigma^2_{z.1}$ and $\sigma^2_{z.2}$ we used the above generated means and the information that the sum of these variances equals $2Var(Z) - \frac{1}{2}(\mu_{z.1} - \mu_{z.2})^2$. Thus, we specified the sub-population variances as follows:

$$\begin{aligned} &\sigma^2_{z.1} = (2Var(Z) - \frac{1}{2}(\mu_{z.1} - \mu_{z.2})^2) \cdot h_2, \\ &\sigma^2_{z.2} = (2Var(Z) - \frac{1}{2}(\mu_{z.1} - \mu_{z.2})^2) \cdot (1 - h_2), \\ &\text{with } h_2 \sim U[0.8, 1]. \end{aligned} \tag{17}$$

The sequential and the fixed ANOVA were then performed on samples from these simulated populations. The remaining simulation settings are as described in the main simulation.

Results

Table 6

Error Rates of the sequential and fixed ANOVA (4 groups, $\alpha = \beta = .05$, N_r = 10,000)

Error Type	f_{true}	f_{exp}	Error Rate Seq (%)[a]	Error Rate Fix (%)[b]	Mean Sequential Sample[c]	Fixed Sample[d]
Type I	**0.00**	0.10	4.4	5.1	1110	1724
		0.25	3.9	4.6	186	280
		0.40	3.4	5	78	112
Type II	**0.10**	0.10	4.9	5.1	973	1724
	0.25	0.10	0	0	275	1724
		0.25	3.2	4.4	170	280
	0.40	0.10	0	0	166	1724
		0.25	0	0	86	280
		0.40	2.4	3.6	72	112

Note: All error rates were not significantly higher than the alpha level of 5% using the lower boundary of the 90% Clopper-Pearson-Interval as criterion; f_{true} = true effect size in the population; f_{exp} = expected effect size; [a] error rate of the sequential ANOVA; [b] error rate of the fixed ANOVA; [c] average sample size of sequential ANOVAs on simulated data; [d] result of an *a priori* power analysis using the *pwr* package in R; N_r = iterations per condition in the simualtion; The table includes only the cases where we have previously shown that the error rates were successfully controlled under ideal conditions ($f_{exp} \leq f_{true}$).

Table 6 shows the error rates and required sample sizes for the sequential and fixed ANOVA applied to the non-normal data. Some error rates of the fixed-sample procedure were descriptively larger than .05, which reflects sampling error, as the lower boundary of

the 90% Clopper-Pearson confidence interval did not exceed the nominal value. Thus, both the sequential ANOVA and the fixed ANOVA control statictical decision error rates even when the normality assumption is violated. Again, however, the sequential ANOVA does so notably more efficiently than the fixed ANOVA.

Estimation of Effect Sizes

SPRTs were developed to increase the efficiency of statistical decision-making while controlling error rates. However, even though hypothesis testing is the main focus in many applications, researchers are typically interested in effect size estimation following the hypothesis test (Fritz et al., 2012; Lovakov & Agadullina, 2021). Thus, discussing the limitations of effect size estimation in the context of a sequential ANOVA is relevant.

Let us assume that researchers want to estimate Cohen's f from a sample that they collected using a sequential ANOVA and that they stopped the data collection process as soon as a decision was reached. Like in a fixed-sample setting, one could calculate Cohen's f and a 95% Confidence Interval (CI). However, several issues should be considered.

First, Cohen's f is known to be a biased estimate in fixed-sample applications (Lakens, 2013; Maxwell et al., 2017). Thus, a better estimate is given by (Maxwell et al., 2017, p. 150):

$$\hat{f}_{\text{adjusted}} = \sqrt{\frac{k-1}{N}\left(\frac{N-k-2}{N-k}F - 1\right)} \tag{18}$$

Second, the high sample size efficiency of sequential ANOVAs leads to smaller sample sizes on average. As is well known, smaller sample sizes are generally less precise in estimating population parameters than larger samples (Gigerenzer, 1989). Thus, there is a trade-off between a high sample size efficiency (leading to smaller samples on average) and estimation precision (requiring larger samples).

In addition, the use of sequential stopping rules produces biased estimates

depending on the sample size. This phenomenon is not only known in the context of SPRTs but sequential procedures in general, such as group sequential designs (Pocock & Hughes, 1989) and the Sequential Bayes Factor Test (Schönbrodt et al., 2017). Figure 3 shows the link between the sample size and the estimated effect size in our simulations for different scenarios. Panel A shows the case of a large population effect size ($f_{\text{true}} = 0.40$). For small sample sizes and correct decisions (i.e., H_0 rejected), the estimates show an upward bias. The expected effect size also influenced the bias: The larger f_{exp}, the larger the bias. For larger samples and correct rejections of H_0, in contrast, the effect size was underestimated. The same pattern emerged when H_0 was incorrectly maintained: In this case, the effect was generally underestimated.

Panel B shows the case where there is no effect in the population. Here, overestimation of the effect size occured particularly when the expected effect was large or medium; thus, when sample sizes were rather small. Overall, these results indicate that small samples tend to overestimate the true effect size and large samples rather underestimate the true effect size.

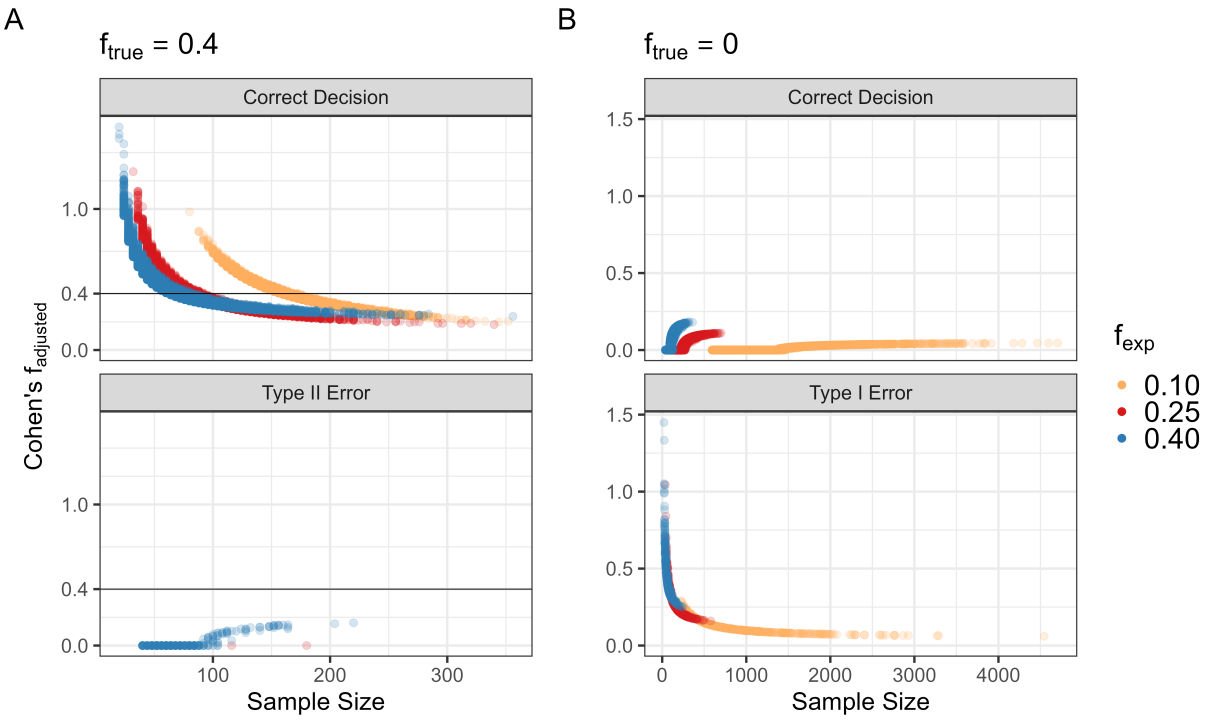
From a meta-analytic viewpoint, however, this bias is not generally problematic. A meta-analytic weighted average of estimated effect sizes across all simulated samples (using $\hat{f}_{\text{adjusted}}$) was essentially unbiased (s. Table A1 in the Appendix for more information). This finding replicates results from Schönbrodt et al. (2017), who showed similar results for meta-analytic effect size estimation using the SBFT.

Finally, a good strategy to calculate confidence intervals for effect size estimates following a sequential test is an ongoing topic of discussion (Bhattacharjee & Mukhopadhyay, 2015; Ulrich & Miller, 2021). A naive approach would use the same calculation method that is used for fixed samples; however, the coverage rate of these confidence intervals is typically lower than the nominal value. Across simulation conditions, we found that the 95% CI calculated for the effect sizes estimated from our simulated

sequential samples covered the population effect size in 93% of the cases. Thus, if unbiased effect size estimation is the primary aim of the study, a sequential testing approach may not be the optimal choice.

Figure 3

Effect Size Estimation (4 groups, $\alpha = \beta = .05$, N_r = 10,000)



Discussion

In this paper, we presented a sequential ANOVA based on Abraham Wald's (1945) SPRT to increase the efficiency of statistical decision making and demonstrated its usefulness in a series of simulation studies. The SPRT is based on the nominal standard for hypothesis testing in psychology, namely, the Neyman-Pearson framework (1933), and it was designed to control decision errors in a maximally efficient way. It uses a likelihood ratio (LR) as a test statistic which indicates the relative evidence for one over a competing hypothesis. LRs have the advantage that they are easily interpretable (e.g., $LR = 20$ means that the data are 20 times more likely under one hypothesis than under the other

hypothesis) and they can be used to quantify evidence for the null as well as for the alternative hypothesis. In addition, SPRTs exert full control over α and β error rates and can be applied to any testing situation as long as the LR is fully specified.

Although SPRTs have many benefits and have existed in the statistics literature for nearly 80 years, they are not widely known and rarely considered part of the methodological toolbox of psychological researchers. We argue that SPRTs can be a powerful and efficient alternative to fixed sample designs and a major asset in overcoming the problem of underpowered studies (Button et al., 2013; Rossi, 1990; Szucs & Ioannidis, 2017). Especially research with limited resources, such as neuromiaging research where studies are often costly in terms of monetary and time resources, can benefit from the high efficiency of SPRTs. The SPRT also caters to moral considerations: A clinical study, for example, where participants in a control group may receive an inferior treatment, has an ethical obligation to use as few participants as possible, that is, make use of the most efficient statistical procedure. Moreover, as the SPRT represents a balanced decision-making process between two hypotheses, its application increases the chance that null effects (i.e., decisions in favor of H_0) are published, thus helping to reduce the notorious *file drawer problem* (Rosenthal, 1979). To enhance hypothesis testing efficiency in the mentioned cases, we implemented a sequential ANOVA based on the SPRT in the R package *sprtt*. We conducted a comprehensive analysis of its properties, comparing them directly to a fixed-sample ANOVA.

Simulation Results

In our main simulation study, we showed that in most of the simulated cases, the sequential sample sizes were smaller than the fixed sample sizes: On average, less than half of the data needed to be collected using a sequential instead of a fixed ANOVA. The sequential ANOVA is especially efficient if the expected effect size is small, allowing for a more conservative and realistic specification of the expected effect size.

Robustness analyses showed similar properties between the sequential and the fixed ANOVA in terms of error probability control, while the sequential test was consistently more efficient than the conventional test even when basic assumptions of the tests were violated. Our results replicate other simulation studies that investigated the robustness of sequential *t*-tests (Lee & Fung, 1980; Schnuerch & Erdfelder, 2020) or fixed ANOVAs (Lix et al., 1996; Schmider et al., 2010), further supporting the notion that sequential tests provide a useful alternative to conventional test procedures.

Limitations of Sequential ANOVAs

Despite their favorable properties, we do not promote sequential ANOVAs as the optimal choice under all circumstances. The selection of an appropriate testing procedure requires researchers to weigh the benefits and limitations of various options in their statistical toolbox and carefully choose the option that best fits the problem at hand (Gigerenzer, 2004). The first limitation of SPRTs is that the exact sample size is not known beforehand. When planning a study, one could use information about average sample sizes from simulations to get an idea of the required sample size. However, if the exact sample size must be known beforehand, there are better choices than the SPRT, such as fixed-sample designs or other, truncated sequential tests (e.g., Miller & Ulrich, 2021; Pramanik et al., 2021; Proschan et al., 2006).

Even though the SPRT will terminate with a finite sample size (Wald, 1947), it is not guaranteed that it terminates before the available resources are exhausted. Thus, theoretically, it could happen that data collection has to be stopped before a decision has been reached. In this case, researchers must refrain from making a decision as error rates will not be controlled for this altered procedure. However, there is no reason not to interpret the available statistical evidence provided by the LR at that point (Schnuerch et al., 2022; Schnuerch & Erdfelder, 2020). A possible strategy is thus to specify a maximum sample size for the sequential ANOVA ($N_{\max}$) based on the *a priori* sample size of the

power analysis for a corresponding fixed design. If sampling terminates before reaching $N_{\max}$, a decision can be made accordingly. If $N_{\max}$ is reached before a decision threshold is crossed, in contrast, researchers can stop and report the evidence provided by the LR without making a finite statistical decision. Our simulations showed that the risk is low, however: Across simulation conditions, the sequential procedure terminated in about 90% of the cases before reaching the sample size of the corresponding fixed-sample test.

Another limitation is that effect size estimation following a sequential test may be biased. Small sequential samples tend to overestimate the population effect, and large sequential samples tend to underestimate the population effect. From a meta-analytic perspective that takes into account both early and late terminations at either decision threshold, however, these biases balance each other out (see Schönbrodt et al., 2017). Nevertheless, if the primary goal of a study is to estimate a population effect size as precisely as possible, a sequential testing procedure (of any kind) cannot be recommended. In contrast, if the main goal is to test a statistical hypothesis – is there an effect of interest or not? – a sequential ANOVA is a valid and highly efficient tool.

Consequently, another limitation concerns the calculation of the confidence interval (CI). The naive approach – using the calculation method for fixed samples (Steiger, 2004) – results in a CI that does not include the true effect sizes as often as it should (Ulrich & Miller, 2021). Using a non-parametric calculation procedure like bootstrapping showed even worse results in the simulation study of Bhattacharjee and Mukhopadhyay (2015). Although the naive approach is flawed, the calculation of the naive CI could still be helpful to some researchers to understand effect size estimates of small sequential samples better. Further research is needed to improve the calculation method for CIs in this context.

Conclusion

The sequential ANOVA based on the SPRT proposed herein is an efficient alternative to classical fixed-sample ANOVAs. It exerts strict control of α and β error rates

and requires on average around 50% smaller samples than conventional test procedures. This advantage in sample size efficiency remains even when underlying assumptions are violated, in which case the sequential ANOVA is similarly robust as the fixed-sample ANOVA. By implementing the sequential ANOVA in the user-friendly R package *sprtt* and thoroughly investigating its properties in a range of possible scenarios by means of simulations, we hope to make the sequential procedure more accessible and pave its way into the statistics toolbox of psychological researchers in all subfields.

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Appendix

Effect Size Estimation

Table A1

Weighted Averages of Cohen's f to Estimate the Population Effect Size (4 groups, $\alpha = \beta = .05$, N_r = 10,000)

		sequential ANOVA		fixed ANOVA	
f_{true}	f_{exp}	f	$f_{adjusted}$	f	$f_{adjusted}$
0.00	**0.10**	0.04	0.01	0.04	0.01
	0.25	0.11	0.03	0.10	0.03
	0.40	0.17	0.05	0.15	0.05
0.10	**0.10**	0.11	0.10	0.11	0.10
	0.25	0.14	0.08	0.14	0.08
	0.40	0.19	0.08	0.18	0.08
0.25	**0.10**	0.27	0.24	0.25	0.25
	0.25	0.28	0.24	0.27	0.24
	0.40	0.30	0.23	0.29	0.22
0.40	**0.10**	0.42	0.39	0.40	0.40
	0.25	0.45	0.39	0.41	0.39
	0.40	0.46	0.39	0.43	0.39

Note: N_r = iterations per condition in the simualtion; f_{exp} = expected effect size; f_{true} = true effect size in the population; f = weighted average of Cohens's f; $f_{adjusted}$ = weighted average of Cohens's f adjusted (Maxwell et al., 2017, p. 150).